

Central Illustration

Resting Diastolic Indices, Cardiopulmonary Fitness, and Heart Failure Outcomes in HCM

441

patients
with HCM

Echo-aligned
CPET,
2006–2021

Resting indices

- E/e'**
n = 316
- LAVi**
n = 301
- TRVmax**
n = 253



1 Exercise capacity

Per SD higher index

Lower peak $\dot{V}O_2$
% predicted

- E/e'** -6.0
- LAVi** -3.6
- TRVmax** -6.3

Higher $\dot{V}E/\dot{V}CO_2$ slope
slope units

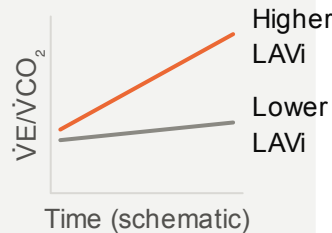
- E/e'** +0.9
- LAVi** +1.0
- TRVmax** +2.5

All 6 associations
 $q \leq 0.045$; linear,
no threshold



2 Change over time

199 patients, serial CPET



**Higher LAVi: steeper
rise in $\dot{V}E/\dot{V}CO_2$ slope**

+0.40 per year per SD
 $q = 0.028$; n = 124

Peak $\dot{V}O_2$ trajectory:
no linear modification
by any index



3 Heart failure events

436 patients, 118 events

Hazard ratio per SD

- LAVi** **1.68**
1.40–2.03
- E/e'** **1.36**
1.17–1.59
- TRVmax** **1.23**
1.02–1.47

- Lower peak $\dot{V}O_2$** **1.74**
1.40–2.16

Median follow-up
4.8 years

Resting diastolic indices relate to exercise capacity and ventilatory efficiency in HCM. Higher LAVi was associated with a steeper rise in $\dot{V}E/\dot{V}CO_2$ slope and with heart failure events; peak $\dot{V}O_2$ remained the strongest single predictor.